

# Code Sample

## Selected Replication Code

Rishav Roy

### CODING SAMPLE: FULL COPY

These excerpts are selected from the replication code for the paper. Selected outputs appear at the end of the document. The [full paper](#) and [repository](#) are available online. The complete paper-facing table and figure assembly code is in [R/output/make\\_tables.R](#) and [R/output/make\\_figures.R](#). This PDF can be generated using `make samples` or `bash scripts/run_full_build.sh`.

### Survey-weighted enrollment model

File: `R/selection/estimate_selection_probit.R`

```
selection_probit_variables <- function(selection_data) {
  controls <- c("AGE", "age", "SEX", "HH_SIZE", "RELIGION", "SOCIAL_GROUP", "SECTOR")
  exclusion_all_kids <- c("DIST_FROM_NEAREST_PRIMARY_CLASS", "father_educ")
  exclusion_district <- c(
    "dmean_num_IS_EDU_FREE", "dmean_num_TUTION_FEE_WAIVED",
    "dmean_num_RECD_SCHOLARSHIP_STIPEND", "dmean_num_RECD_TXT_BOOKS",
    "dmean_num_RECD_STATIONERY", "dmean_num_MID_DAY_MEAL_ETC_RECD",
    "dmean_num_ENROLLMENT_COST"
  )
  intersect(c(controls, exclusion_all_kids, exclusion_district), names(selection_data))
}

# ' project the selection sample to the model's actual dependency surface
# '
# ' Keeping unrelated Block-5 schooling attributes in the fitted survey design makes
# ' the serialized model change whenever those attributes change, even though they
# ' are absent from the probit formula. A narrow model frame lets {targets} skip the
# ' expensive AME branch when upstream changes do not affect estimation.
project_selection_model_data <- function(selection_data) {
  selection_data <- safe_df(selection_data)
  design <- c("enrolled", "FSU_SL_NO", "weight", "STATE", "STRATUM", "SUB_STRATUM_NO")
  keep <- unique(c(design, selection_probit_variables(selection_data)))
  selection_data[intersect(keep, names(selection_data))]
}

# ' estimate selection probit
```

```

#'
estimate_selection_probit <- function(selection_data, cfg) {
  if ("enrolled" %in% names(selection_data) || all(is.na(selection_data$enrolled))) {
    return(list(status = "out_of_active_pipeline", reason = "No_enrolled_variable."))
  }
  covars <- selection_probit_variables(selection_data)
  if (!length(covars)) {
    return(list(status = "out_of_active_pipeline", reason = "No_probit_covariates."))
  }
  f_probit <- stats::reformulate(covars, response = "enrolled")
  if (identical(cfg$mode, "final") && requireNamespace("survey", quietly = TRUE)) {
    old_options <- options(survey.lonely.psu = "average")
    on.exit(options(old_options), add = TRUE)

    design <- build_survey_design_selection(selection_data)
    if (!is.null(design)) {
      out <- fit_selection_probit(design, f_probit)
      return(stabilize_selection_model_formula(out, f_probit))
    }
  }
  out <- stats::glm(f_probit, data = selection_data, family = stats::binomial(link = "probit"))
  stabilize_selection_model_formula(out, f_probit)
}

#' store a durable selection-model formula
#'
#' Programmatically fitted models otherwise retain a call to the local symbol
#' `f_probit`. Packages which reconstruct model data from the saved call cannot
#' resolve that symbol after the model is serialized by targets. Embed the
#' formula object in the call and retain the existing audit attribute.
stabilize_selection_model_formula <- function(model, formula) {
  if (!inherits(formula, "formula")) {
    stop("Selection_model_formula_must_inherit_from_formula.", call. = FALSE)
  }
  if (!is.null(model$call)) model$call$formula <- formula
  attr(model, "selection_probit_formula") <- formula
  model
}

#' build survey design selection
#'
build_survey_design_selection <- function(selection_df) {
  psu <- first_col(selection_df, c("FSU_SL_NO", "fsu", "PSU", "psu"))
  weight <- first_col(selection_df, c("weight", "WEIGHT", "Multiplier", "multiplier"))
  strata_cols <- intersect(c("STATE", "STRATUM", "SUB_STRATUM_NO"), names(selection_df))
  if (is.null(psu) || is.null(weight) || !length(strata_cols)) return(NULL)
  selection_df$survey_strata <- interaction(selection_df[strata_cols], drop = TRUE)
  survey::svydesign(
    ids = stats::as.formula(paste0("~", psu)),

```

```

strata = ~.survey__strata,
weights = stats::as.formula(paste0("~", weight)),
data = selection__df,
nest = TRUE
)
}

#' fit selection probit
#'
fit_selection__probit <- function(selection__design, f__probit) {
  survey::svyglm(f__probit, design = selection__design, family = stats::quasibinomial(link = "probit"))
}

```

## Average marginal effects on the model estimation sample

File: R/selection/compute\_average\_marginal\_effects.R

```

ame__model__weight <- function(model__data, model__weights = NULL) {
  weight__cols <- intersect(c("weight", "WEIGHT", "Multiplier", "multiplier", "(weights)"), names(
    model__data))
  if (length(weight__cols)) {
    return(suppressWarnings(as.numeric(model__data[[weight__cols[[1]]]])))
  }

  if (!is.null(model__weights) && length(model__weights) == nrow(model__data)) {
    return(suppressWarnings(as.numeric(model__weights)))
  }

  NULL
}

#' build marginaleffects newdata and weights
#'
#' @return List with `data`, `wts`, and `has__explicit__weights`.
ame__model__data_and__weights <- function(model) {
  model__data <- as.data.frame(stats::model.frame(model))
  model__weights <- tryCatch(stats::weights(model), error = function(e) NULL)
  weight <- ame__model__weight(model__data, model__weights)

  if (!is.null(weight) && length(weight) == nrow(model__data) && any(is.finite(weight) & weight
    != 1)) {
    model__data$ame__weight <- weight
    return(list(data = model__data, wts = ".ame__weight", has__explicit__weights = TRUE))
  }

  list(data = model__data, wts = FALSE, has__explicit__weights = FALSE)
}

#' run marginaleffects while muffling only a redundant explicit-weight warning

```

```

#'
run_avg_slopes <- function(model, model_data, wts, numberiv = NULL) {
  args <- list(
    model = model,
    newdata = model_data,
    wts = wts,
    vcov = TRUE,
    type = "response"
  )
  if (!is.null(numberiv)) args$numberiv <- numberiv

  withCallingHandlers(
    do.call(marginaleffects::avg_slopes, args),
    warning = function(w) {
      msg <- conditionMessage(w)
      explicit_weight_warning <- grepl(
        "normally good practice to specify weights using the `wts` argument",
        msg,
        fixed = TRUE
      )
      if (explicit_weight_warning && !identical(wts, FALSE)) {
        invokeRestart("muffleWarning")
      }
    }
  )
}

#' compute ames autodiff
#'
compute_ames_autodiff <- function(model) {
  # Use the exact estimation sample retained by glm/svyglm. Passing the full
  # pre-model selection_data can have extra rows omitted during model fitting,
  # which makes marginaleffects recycle weights/covariates silently.
  amed <- ame_model_data_and_weights(model)
  run_avg_slopes(model, amed$data, amed$wts)
}

```

## Montiel Olea-Pflueger effective-F calculation

File: R/iv/weak\_identification.R

```

mop_effective_f <- function(
  model, analysis_data, tau = 0.10, size = 0.05) {
  unavailable <- function(reason) {
    list(
      statistic = NA_real_, critical_value = NA_real_, p.value = NA_real_,
      effective_df = NA_real_, tau = tau, size = size,
      status = "not_estimated", reason = reason
    )
  }
}

```

```

}

if (linherits(model, "ivreg")) {
  return(unavailable("Montiel_Olea-Pflueger_effective_F_requires_an_ivreg_model."))
}
if (!requireNamespace("momentfit", quietly = TRUE)) {
  return(unavailable("Package 'momentfit' is not installed."))
}
if (!is.finite(tau) || tau <= 0 || !is.finite(size) || size <= 0 || size >= 1) {
  return(unavailable("Montiel_Olea-Pflueger_tau_and_size_must_be_valid_probabilities."))
}

terms <- parse_iv_formula_terms(model)
if (is.null(terms)) {
  return(unavailable("Could not parse IV formula for Montiel_Olea-Pflueger_effective_F."))
}
endogenous <- setdiff(terms$regressors, terms$instruments)
if (length(endogenous) != 1L) {
  return(unavailable("Montiel_Olea-Pflueger_effective_F_requires_exactly_one_endogenous_regressor."
  ))
}

formula <- stats::formula(model)
regression_formula <- stats::as.formula(
  call("~", formula[[2L]], formula[[3L]][[2L]]),
  env = environment(formula)
)
instrument_formula <- stats::as.formula(
  call("~", formula[[3L]][[3L]]),
  env = environment(formula)
)
needed <- unique(c(all.vars(regression_formula), all.vars(instrument_formula)))
data <- iv_analysis_frame(analysis_data, needed)
missing <- setdiff(needed, names(data))
if (length(missing)) {
  return(unavailable(paste(
    "Missing variables for Montiel_Olea-Pflueger_effective_F:",
    paste(missing, collapse = ", ")
  )))
}
moment_data <- data[needed]
if (any(!stats::complete.cases(moment_data))) {
  return(unavailable("Montiel_Olea-Pflueger_analysis_data contain incomplete model rows."))
}

stored_cluster <- attr(model, "cluster_state", exact = TRUE)
cluster <- if (
  length(stored_cluster) == stats::nobs(model) &&
  !anyNA(stored_cluster) && length(unique(stored_cluster)) >= 2L
) {

```

```

    as.vector(stored__cluster)
  } else {
    iv__model__cluster(model, data)
  }
  covariance <- if (is.null(cluster)) "MDS" else "CL"
  covariance__options <- if (is.null(cluster)) {
    list(type = "HC0")
  } else {
    # momentfit's clustered moment covariance is based on sandwich::meatCL()
    # and supports HC0. Keep its package-native finite-cluster correction
    # rather than altering the existing HC1 Wald-F diagnostic to force equality.
    list(
      cluster = data.frame(cluster = cluster),
      type = "HC0", cadjust = TRUE, multi0 = FALSE
    )
  }
}

result <- tryCatch({
  moment__model <- momentfit::momentModel(
    regression__formula,
    instrument__formula,
    data = moment__data,
    vcov = covariance,
    vcovOptions = covariance__options
  )
  momentfit::MOPtest(
    moment__model,
    tau = tau,
    size = size,
    estMethod = "TSLS",
    simplified = TRUE,
    print = FALSE
  )
}, error = function(e) e)
if (inherits(result, "error")) return(unavailable(conditionMessage(result)))

values <- suppressWarnings(as.numeric(result[c("Feff", "critValue", "pvalue", "Keff"])))
names(values) <- c("statistic", "critical_value", "p.value", "effective_df")
if (length(values) != 4L || any(!is.finite(values))) {
  return(unavailable("momentfit::MOPtest() did not return finite effective-F diagnostics."))
}

c(
  as.list(values),
  list(tau = tau, size = size, status = "estimated", reason = NA__character__)
)
}

```

## Anderson–Rubin inference and bounded exclusion sensitivity

File: R/iv/weak\_identification.R

```
anderson_rubin_test <- function(
  data, outcome, treatment, excluded, included = character(),
  controls = character(), fixed_effect = "none", cluster, beta0 = 0
) {
  transformed <- ".ar_outcome"
  data[[transformed]] <- num(data[[outcome]]) - beta0 * num(data[[treatment]])
  rhs <- unique(c(excluded, included, controls, iv_fixed_effect_terms(fixed_effect)))
  fit <- stats::lm(stats::reformulate(rhs, response = transformed), data = data)
  test <- clustered_joint_wald_test(fit, excluded, cluster)
  c(statistic = test[["statistic"]], p.value = test[["p.value"]])
}

# Weak-IV-robust exclusion sensitivity for one scalar excluded instrument.
# For a candidate structural effect beta, regress Y - beta D on Z and the same
# nuisance terms used by the registered IV specification. The coefficient on Z
# is the direct effect gamma that would reconcile that beta with the data.
# Subtracting any fixed gamma * Z changes that coefficient but not the residuals,
# so the clustered standard error is invariant to gamma. This makes the union of
# Anderson--Rubin confidence sets over gamma in [lower, upper] available without
# a numerical gamma grid.
bounded_exclusion_ar_profile <- function(
  data, outcome, treatment, excluded, included = character(),
  controls = character(), fixed_effect = "none", cluster, beta_values) {
  excluded <- plain_chr(excluded)
  if (length(excluded) != 1L || is.na(excluded[[1L]]) || !nzchar(excluded[[1L]])) {
    stop("Bounded_exclusion_sensitivity_requires_exactly_one_excluded_instrument.", call. = FALSE)
  }
  beta_values <- num(beta_values)
  if (!length(beta_values) || any(!is.finite(beta_values))) {
    stop("Bounded_exclusion_sensitivity_requires_finite_beta_values.", call. = FALSE)
  }
  instrument <- excluded[[1L]]
  rhs <- unique(c(instrument, included, controls, iv_fixed_effect_terms(fixed_effect)))

  safe_bind_rows(lapply(beta_values, function(beta0) {
    transformed <- ".bounded_exclusion_ar_outcome"
    x <- as.data.frame(data)
    x[[transformed]] <- num(x[[outcome]]) - beta0 * num(x[[treatment]])
    fit <- stats::lm(stats::reformulate(rhs, response = transformed), data = x)
    inference <- iv_clustered_inference(fit, cluster)
    term <- model_term_inference(fit, instrument, inference$vcov)
    data.frame(
      beta = beta0,
      direct_effect_estimate = term[["estimate"]],
      direct_effect_std.error = term[["std.error"]],
      residual_df = stats::df.residual(fit),

```

```

status = if (
  identical(inference$status, "estimated") &&
  all(is.finite(term[c("estimate", "std.error"]))) &&
  term[["std.error"]] > 0
) "estimated" else "inference__unavailable",
reason = if (identical(inference$status, "unavailable")) inference$reason else NA__character__,
stringsAsFactors = FALSE
)
})))
}

```

## Linked state-sector price series

File: R/prices/build\_temporal\_price\_series.R

```

build_temporal_price_series <- function(
  price_sources,
  switch_date = as.Date("2013-01-01"),
  overlap_start = as.Date("2013-01-01"),
  overlap_end = as.Date("2014-12-01"),
  minimum_link_months = 6L,
  pre_switch_start = NULL,
  pre_switch_end = NULL) {
  switch_date <- price_boundary(switch_date)
  old <- select_pre_2013_price_series(price_sources)
  new <- select_post_2013_price_series(price_sources)
  links <- summarise_price_links(old, new, overlap_start, overlap_end)

  minimum_link_months <- as.integer(minimum_link_months)
  if (length(minimum_link_months) != 1L || is.na(minimum_link_months) || minimum_link_months < 1L) {
    stop("minimum_link_months must be one positive integer.", call. = FALSE)
  }

  old_groups <- unique(old[old$period < switch_date, c("state_code", "sector"), drop = FALSE])

  linked_groups <- links[links$link_months >= minimum_link_months, c("state_code", "sector"),
    drop = FALSE]
  linked_key <- paste(linked_groups$state_code, linked_groups$sector, sep = "\r")
  missing_key <- !paste(old_groups$state_code, old_groups$sector, sep = "\r") %in% linked_key
  if (any(missing_key)) {
    bad <- old_groups[missing_key, , drop = FALSE]
    stop(
      "Pre-2013 price series lack a sufficient direct link to CPI-R/U:",
      paste(paste(bad$state_code, bad$sector, sep = "/"), collapse = ", "),
      call. = FALSE
    )
  }
}

```

```

old <- merge(old, links, by = c("state_code", "sector"), all.x = TRUE, sort = FALSE)
old <- old[old$period < switch_date, , drop = FALSE]
if (!is.null(pre_switch_start) || !is.null(pre_switch_end)) {
  if (is.null(pre_switch_start) || is.null(pre_switch_end)) {
    stop("Both pre_switch_start and pre_switch_end are required when trimming the old price series.", call. = FALSE)
  }
  pre_switch_start <- price_boundary(pre_switch_start)
  pre_switch_end <- price_boundary(pre_switch_end)
  if (pre_switch_end < pre_switch_start || pre_switch_end >= switch_date) {
    stop("The pre-switch production window must end before the switch date.", call. = FALSE)
  }
  old <- old[old$period >= pre_switch_start & old$period <= pre_switch_end, , drop = FALSE]
}
old$index_unlinked <- old$index
old$index <- num(old$index) * num(old$link_factor)

new <- new[new$period >= switch_date, , drop = FALSE]
new$link_factor <- 1
new$link_months <- NA_integer_
new$link_ratio_mad <- NA_real_
new$link_ratio_min <- NA_real_
new$link_ratio_max <- NA_real_
new$index_unlinked <- new$index

columns <- c(
  "state_code", "sector", "period", "index", "index_unlinked", "price_source",
  "link_factor", "link_months", "link_ratio_mad", "link_ratio_min", "link_ratio_max"
)
index <- rbind(old[columns], new[columns])
index$year <- as.integer(format(index$period, "%Y"))
index$month <- as.integer(format(index$period, "%m"))
index <- index[order(index$state_code, index$sector, index$period), , drop = FALSE]
rownames(index) <- NULL
validate_temporal_price_chain(index, switch_date)

structure(
  list(index = index, links = links, switch_date = switch_date),
  class = "emi_temporal_price_series"
)
}

```

## Cross-vintage district candidate scoring

File: R/districts/lineage\_matching.R

```

score_match_candidates <- function(source_roster, reference_units, excluded_source_ids =
  character()) {

```

```

need_pkg("stringdist", "district-lineage_candidate_scores")
source_roster <- safe_df(source_roster)
reference_units <- safe_df(reference_units)
if (!"raw_district" %in% names(source_roster)) source_roster$raw_district <- source_roster$
  district_std
source_open <- source_roster[!source_roster$source_row_id %in% excluded_source_ids, ,
  drop = FALSE]
if (!nrow(source_open)) return(empty_source_candidates())

pairs <- merge(
  source_open, reference_units,
  by = "state_std", all = FALSE, sort = FALSE,
  suffixes = c("__source", "__candidate")
)
if (!nrow(pairs)) return(empty_source_candidates())
source_name <- pairs$district_std_source
candidate_name <- pairs$district_std_candidate
pairs$jw <- 1 - stringdist::stringdist(source_name, candidate_name, method = "jw", p = 0.1)
dl_distance <- stringdist::stringdist(source_name, candidate_name, method = "dl")
denom <- pmax(nchar(source_name), nchar(candidate_name), 1L)
pairs$dl <- pmax(0, 1 - dl_distance / denom)
pairs$trigram <- stringdist::stringsim(source_name, candidate_name, method = "cosine", q = 3)
pairs$token <- token_jaccard_similarity(source_name, candidate_name)
pairs$score <- rowMeans(pairs[c("jw", "dl", "trigram", "token")], na.rm = TRUE)

ranked <- safe_bind_rows(lapply(split(seq_len(nrow(pairs)), pairs$source_row_id),
  function(i) {
    x <- pairs[i, , drop = FALSE]
    preference <- vintage_preference(x$wave[[1]])
    vintage_rank <- match(x$reference_vintage, preference, nomatch = length(preference) + 1L)
    x <- x[order(-x$score, -x$jw, vintage_rank, x$unit_id), , drop = FALSE]
    x$rank <- match(x$district_std_candidate, unique(x$district_std_candidate))
    distinct_scores <- x$score[!duplicated(x$district_std_candidate)]
    second <- if (length(distinct_scores) >= 2L) distinct_scores[[2]] else -Inf
    x$margin <- ifelse(x$rank == 1L, x$score - second, NA_real_)
    x$preferred_vintage <- !duplicated(x$district_std_candidate)
    x
  })

candidate_groups <- split(seq_len(nrow(ranked)), ranked$unit_id)
best_by_candidate <- stats::setNames(
  vapply(candidate_groups, function(i) {
    score <- ranked$score[i]
    ranked$source_row_id[i[which.max(score)]]
  }, character(1)),
  names(candidate_groups)
)
ranked$reciprocal_nearest <- ranked$rank == 1L &
  unname(best_by_candidate[ranked$unit_id] == ranked$source_row_id)
threshold <- district_match_candidate_thresholds()

```

```

ranked$high_precision_candidate <- ranked$rank == 1L & ranked$preferred_vintage &
  ranked$reciprocal_nearest &
  directional_tokens_compatible(ranked$district_std_source, ranked$district_std_candidate) &
  ranked$jw >= threshold[["jw"]] & ranked$dl >= threshold[["dl"]] &
  ranked$trigram >= threshold[["trigram"]] & ranked$margin >= threshold[["margin"]]
ranked <- ranked[ranked$rank <= 5L, , drop = FALSE]

data.frame(
  source_row_id = ranked$source_row_id,
  wave = ranked$wave,
  source_code = ranked$source_code,
  state_std = ranked$state_std,
  source_name_raw = ranked$raw_district,
  source_name = ranked$district_std_source,
  candidate_unit = ranked$unit_id,
  candidate_name = ranked$district_std_candidate,
  candidate_source_id = ranked$source_id,
  reference_vintage = ranked$reference_vintage,
  candidate_method = "fuzzy_name_candidate",
  rank = ranked$rank,
  jw = ranked$jw,
  dl = ranked$dl,
  trigram = ranked$trigram,
  token = ranked$token,
  score = ranked$score,
  margin = ranked$margin,
  reciprocal_nearest = ranked$reciprocal_nearest,
  high_precision_candidate = ranked$high_precision_candidate,
  stringsAsFactors = FALSE
)
}

```

## Selected Outputs

The tables and figures below are generated by the analyses from which the preceding excerpts are drawn.

### Selected enrollment-model average marginal effects

Selected average slopes and discrete comparisons from the survey-weighted enrollment model.

| Covariate or comparison                    | Estimate | SE    |
|--------------------------------------------|----------|-------|
| Religion: Muslim (ref: Hindu)              | -0.125   | 0.013 |
| Social group: Scheduled Tribe (ref: Other) | -0.101   | 0.019 |
| Social group: Scheduled Caste              | -0.074   | 0.015 |
| Social group: Other Backward Class         | -0.029   | 0.012 |
| Distance 1–2km (ref: <1km)                 | -0.076   | 0.044 |

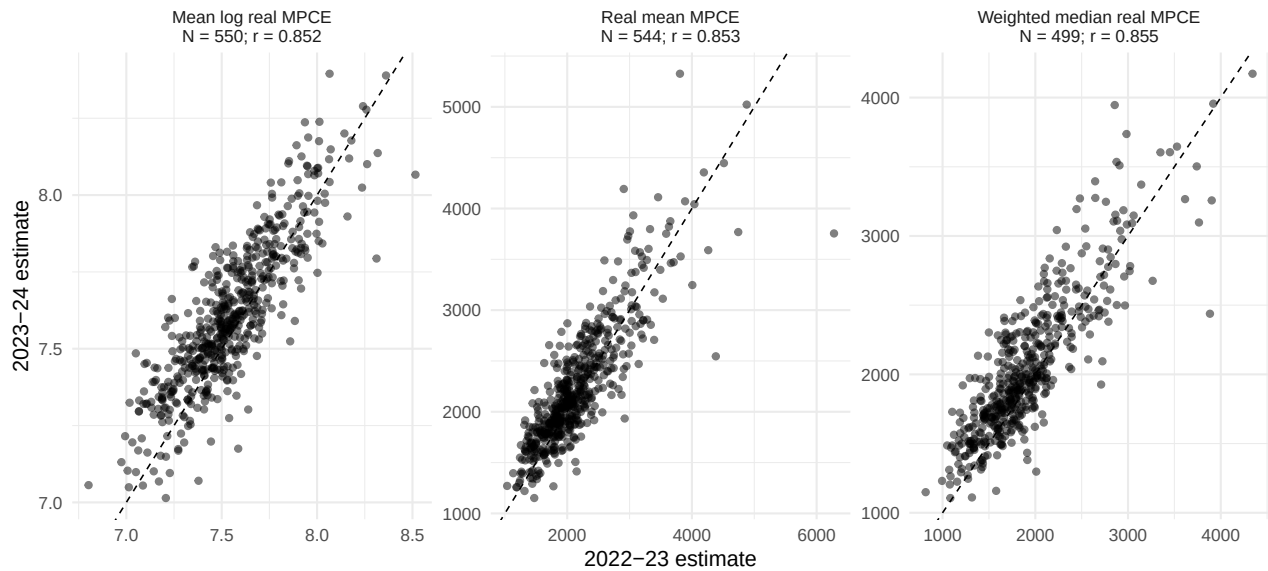
Weak-instrument-robust long-run inference

Long-run IV estimates, effective first-stage F statistics, and Anderson–Rubin acceptance sets.

| Outcome             | 2SLS estimate | SE    | Effective F | AR 95% set                                                    |
|---------------------|---------------|-------|-------------|---------------------------------------------------------------|
| 2022-23 long change | 0.117         | 0.152 | 0.686       | [grid<= -0.193146, -0.0714641] U [0.025109, 0.193146 <=grid]  |
| 2023-24 long change | 0.137         | 0.234 | 0.414       | [grid<= -0.185982, -0.0511449] U [0.0148785, 0.185982 <=grid] |

Consumption-series consistency

Validation of the harmonized consumption series used in the later-outcome analysis.



District-lineage construction

Summary of the cross-vintage district-lineage logic used to harmonize changing district boundaries.

## Appendix A2. District-lineage logic

Only deterministic, adjudicated mappings enter the primary Census-2001 analytical geography

